

Berezin Integration and the Gross-Neveu Model

Perimeter Scholars International, Quantum Field Theory II, Tutorial 7; 10 worked parts.

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Quantum Field Theory

Question prompts paraphrased from the official Perimeter PSI Online sheet; the equations and notation follow the source.

1. Grassmann Variables

Let θ_a and $\bar{\theta}_a$, with $a = 1, \dots, N$, be N conjugate pairs of Grassmann variables, and let A be an $N \times N$ matrix. Begin from the identity

$$\int \prod_{a=1}^N d\bar{\theta}_a d\theta_a \exp \left[- \sum_{c,d} \bar{\theta}_c A_{cd} \theta_d \right] = \det(A). \quad (1.1)$$

- a) Introduce independent conjugate Grassmann sources $\bar{\eta}_c$ and η_c , which anticommute with one another and with every θ_a and $\bar{\theta}_a$. Show that

$$\begin{aligned} Z[\bar{\eta}, \eta] &= \int \prod_{a=1}^N d\bar{\theta}_a d\theta_a \exp \left[- \sum_{c,d} \bar{\theta}_c A_{cd} \theta_d + \sum_c (\bar{\eta}_c \theta_c + \bar{\theta}_c \eta_c) \right] \\ &= \det(A) \exp \left[\sum_{c,d} \bar{\eta}_c (A^{-1})_{cd} \eta_d \right]. \end{aligned} \quad (1.2)$$

Solution. We shift the integral measure by

$$\tilde{\theta}_a = \theta_a - A_{ab}^{-1} \eta_b. \quad (1.3)$$

Then, the exponent becomes

$$-(\bar{\tilde{\theta}}_c + \bar{\eta}_b A_{bc}^{-1}) A_{cd} (\tilde{\theta}_d + A_{da}^{-1} \eta_a) + \bar{\eta}_c (\tilde{\theta}_c + A_{ca}^{-1} \eta_a) + (\bar{\tilde{\theta}}_c + \bar{\eta}_b A_{bc}^{-1}) \eta_c = -\bar{\tilde{\theta}}_c A_{cd} \tilde{\theta}_d + \bar{\eta}_c A_{cd}^{-1} \eta_d, \quad (1.4)$$

which follows

$$Z[\bar{\eta}, \eta] = \det A \exp(\bar{\eta}_c A_{cd}^{-1} \eta_d). \quad (1.5)$$

- b) Use $Z[\bar{\eta}, \eta]$ and the differentiation rules for Grassmann variables to evaluate

$$\langle \theta_a \bar{\theta}_b \rangle = \frac{\int \prod_c d\bar{\theta}_c d\theta_c e^{-\bar{\theta} \cdot A \cdot \theta} \theta_a \bar{\theta}_b}{\int \prod_d d\bar{\theta}_d d\theta_d e^{-\bar{\theta} \cdot A \cdot \theta}}, \quad \langle \theta_a \theta_b \rangle. \quad (1.6)$$

Solution. Using

$$\frac{\partial Z[\bar{\eta}, \eta]}{\partial \bar{\eta}_a} = \int \prod_{a=1}^N d\bar{\theta}_a d\theta_a \theta_a \exp \left[- \sum_{c,d} \bar{\theta}_c A_{cd} \theta_d + \sum_c (\bar{\eta}_c \theta_c + \bar{\theta}_c \eta_c) \right], \quad (1.7)$$

and

$$-\frac{\partial Z[\bar{\eta}, \eta]}{\partial \eta_b} = \int \prod_{a=1}^N d\bar{\theta}_a d\theta_a \bar{\theta}_b \exp \left[- \sum_{c,d} \bar{\theta}_c A_{cd} \theta_d + \sum_c (\bar{\eta}_c \theta_c + \bar{\theta}_c \eta_c) \right]. \quad (1.8)$$

We can write

$$\begin{aligned}
 \langle \theta_a \bar{\theta}_b \rangle &= -\frac{1}{Z[0,0]} \frac{\partial^2}{\partial \bar{\eta}_a \partial \eta_b} Z[\bar{\eta}, \eta] \Big|_{\eta=0} \\
 &= -\frac{\partial^2}{\partial \bar{\eta}_a \partial \eta_b} \exp(\bar{\eta}_c A_{cd}^{-1} \eta_d) \Big|_{\eta=0} \\
 &= \frac{\partial}{\partial \bar{\eta}_a} [\bar{\eta}_c A_{cd}^{-1} \exp(\bar{\eta}_c A_{cd}^{-1} \eta_d)] \Big|_{\eta=0} \\
 &= A_{cd}^{-1}.
 \end{aligned} \tag{1.9}$$

c) Establish

$$\langle \theta_a \bar{\theta}_b \theta_c \bar{\theta}_d \rangle = (A^{-1})_{ab} (A^{-1})_{cd} - (A^{-1})_{ad} (A^{-1})_{cb}, \tag{1.10}$$

and explain how the two pairings and their relative sign realize Wick's theorem for fermions.

Solution. Using the same method, we have

$$\begin{aligned}
 \langle \theta_a \bar{\theta}_b \theta_c \bar{\theta}_d \rangle &= -\frac{\partial^2}{\partial \bar{\eta}_a \partial \eta_b} \left[A_{cd}^{-1} - \bar{\eta}_e A_{ed}^{-1} A_{cf}^{-1} \eta_f \exp(\bar{\eta}_c A_{cd}^{-1} \eta_d) \right] \Big|_{\eta=0} \\
 &= A_{ab}^{-1} A_{cd}^{-1} - A_{ad}^{-1} A_{cb}^{-1}.
 \end{aligned} \tag{1.11}$$

In terms of Wick's theorem, this is

$$\begin{aligned}
 \langle \theta_a \bar{\theta}_b \theta_c \bar{\theta}_d \rangle &= \overline{\theta_a \bar{\theta}_b} \overline{\theta_c \bar{\theta}_d} + \overline{\theta_a \bar{\theta}_c} \overline{\theta_b \bar{\theta}_d} \\
 &= \overline{\theta_a \bar{\theta}_b} \overline{\theta_c \bar{\theta}_d} - \overline{\theta_a \bar{\theta}_d} \overline{\theta_b \bar{\theta}_c} \\
 &= A_{ab}^{-1} A_{cd}^{-1} - A_{ad}^{-1} A_{cb}^{-1}.
 \end{aligned} \tag{1.12}$$

Now take the continuum limit $\theta_a \rightarrow \psi_\alpha(x)$, where $\alpha = 1, 2, 3, 4$ is a suppressed Dirac index, and replace A_{cd} by the integral kernel $A(x, y)$. Use

$$\{\psi(x), \psi(y)\} = 0, \quad \frac{\delta \psi(x)}{\delta \psi(y)} = \delta^4(x - y). \tag{1.13}$$

d) In the finite-dimensional case, prove

$$\left\{ \frac{d}{d\theta_a}, \theta_b \right\} = \delta_{ab}, \quad \left\{ \frac{d}{d\theta_a}, \frac{d}{d\theta_b} \right\} = 0, \tag{1.14}$$

and write the corresponding functional relations for the Grassmann fields $\psi(x)$.

Solution.

$$\left\{ \frac{d}{d\theta_a}, \theta_b \right\} f(\theta) = \delta_{ab} f(\theta) - \theta_b \frac{df}{d\theta_a} + \theta_b \frac{df}{d\theta_a} = \delta_{ab} f(\theta) \quad \forall f(\theta). \tag{1.15}$$

$$\left\{ \frac{d}{d\theta_a}, \frac{d}{d\theta_b} \right\} = \frac{d}{d\theta_a} \frac{d}{d\theta_b} + \frac{d}{d\theta_b} \frac{d}{d\theta_a} = 0. \tag{1.16}$$

e) Complete the continuum Gaussian identity

$$\begin{aligned}
 \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left[- \int d^4 x' d^4 x \bar{\psi}(x') A(x', x) \psi(x) \right. \\
 \left. + \int d^4 x (\bar{\psi}(x) \eta(x) + \bar{\eta}(x) \psi(x)) \right] = \dots
 \end{aligned} \tag{1.17}$$

Solution. We can read this off directly from the finite version,

$$Z[\bar{\eta}, \eta] = \det A'(x, x') \exp \left(\int d^d x \int d^d x' \bar{\eta}(x) A'^{-1}(x, x') \eta(x') \right). \quad (1.18)$$

f) Determine the two-point correlation function when

$$A = i\gamma \cdot \partial - m. \quad (1.19)$$

Solution.

$$\langle \psi(x) \bar{\psi}(y) \rangle = A'^{-1}(x, y) = \int \frac{d^d k}{(2\pi)^d} \frac{-e^{ik \cdot (x-x')}}{k + m}. \quad (1.20)$$

2. Gross–Neveu Model

Consider the two-dimensional Gross–Neveu model

$$\mathcal{L}_{\text{GN}} = \bar{\psi}_a i \not{\partial} \psi^a + \frac{g^2}{2N} (\bar{\psi}_a \psi^a)^2, \quad (2.1)$$

where $a = 1, \dots, N$ is a colour index and each suppressed Dirac index has two values.

a) Find the mass dimension of g .

Solution. $[g] = 0$.

b) Choose

$$\gamma^0 = \sigma^2, \quad \gamma^1 = i\sigma^1, \quad \gamma^5 = \gamma^0 \gamma^1 = \sigma^3. \quad (2.2)$$

Show that $\mathcal{L}_{\text{GN}}$ is invariant under $\psi_a \mapsto \gamma^5 \psi_a$, and explain why this symmetry excludes a fermion mass term.

Solution. Under the transformation

$$\bar{\psi} \gamma^\mu \psi \rightarrow \psi^\dagger \gamma^5 \gamma^0 \gamma^\mu \gamma^5 \psi = \bar{\psi} \gamma^\mu \psi. \quad (2.3)$$

This can be checked using explicit matrix multiplication. Meanwhile,

$$\bar{\psi} \psi = \psi^\dagger \gamma^0 \psi \rightarrow \psi^\dagger \gamma^5 \gamma^0 \gamma^5 \psi = -\psi^\dagger \gamma^0 \psi = -\bar{\psi} \psi. \quad (2.4)$$

Therefore, the Dirac mass is not allowed, and

$$(\bar{\psi} \psi)^2 \rightarrow (\bar{\psi} \psi)^2. \quad (2.5)$$

c) Show that the model is equivalent to the auxiliary-scalar theory

$$\mathcal{L}_{\psi, \sigma} = \bar{\psi}_a (i \not{\partial} - \sigma) \psi^a - \frac{N}{2g^2} \sigma^2, \quad (2.6)$$

where $\sigma(x)$ is a scalar field. *Hint: perform the functional integral over σ .*

Solution. Integrating over the scalar field, we have

$$\begin{aligned}
& \int \mathcal{D}\sigma \exp \left(- \int d^2x \bar{\psi}_a (i\cancel{\partial} - \sigma) \psi^a - \frac{N}{2g^2} \sigma^2 \right) \\
&= \exp \left(- \int d^2x \bar{\psi}_a i\cancel{\partial} \psi \right) \int \mathcal{D}\sigma \exp \left(- \int d^2x \left(\frac{N}{2} \frac{1}{g} \sigma + \frac{g}{\sqrt{2N}} (\bar{\psi}_a \psi^a) \right)^2 + \frac{g^2}{2N} (\bar{\psi}_a \psi^a)^2 \right) \quad (2.7) \\
&= \text{const} \times \exp \left(- \int d^2x \mathcal{L}_{\text{GN}} \right).
\end{aligned}$$

d) Instead perform the functional integral over ψ and show that the model becomes the scalar-field theory

$$\mathcal{L}_\sigma = -\frac{N}{2g^2} \sigma^2 + N \text{tr} \log(i\cancel{\partial} - \sigma). \quad (2.8)$$

Solution. Integrating over the fermion fields, we have

$$\begin{aligned}
& \int \mathcal{D}\bar{\psi}_a \mathcal{D}\psi_a \exp \left(- \int d^2x \bar{\psi}_a (i\cancel{\partial} - \sigma) \psi^a - \frac{N}{2g^2} \sigma^2 \right) \\
&= \exp \left(- \int d^2x - \frac{N}{2g^2} \sigma^2 \right) \left(\int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left(- \int d^2x \bar{\psi} (i\cancel{\partial} - \sigma) \psi \right) \right)^N \quad (2.9) \\
&= \exp \left(- \int d^2x - \frac{N}{2g^2} \sigma^2 \right) \det(i\cancel{\partial} - \sigma)^N \\
&= \exp \left(- \int d^2x - \frac{N}{2g^2} \sigma^2 + N \text{tr} \log(i\cancel{\partial} - \sigma) \right).
\end{aligned}$$